MACHINE LEARNING INTERNSHIP

TASK 2: HOUSE PRICE PREDICTION

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Colab Notebook: Task 2 Internspark ML.ipynb

1. PROJECT OVERVIEW

The objective of this project is to develop a machine learning regression model capable of predicting house prices based on various property characteristics. The project focuses on exploratory data analysis, feature engineering, data preprocessing, model training, model comparison, and performance evaluation.

2. DATASET INFORMATION

Dataset Name: House Price Prediction Dataset

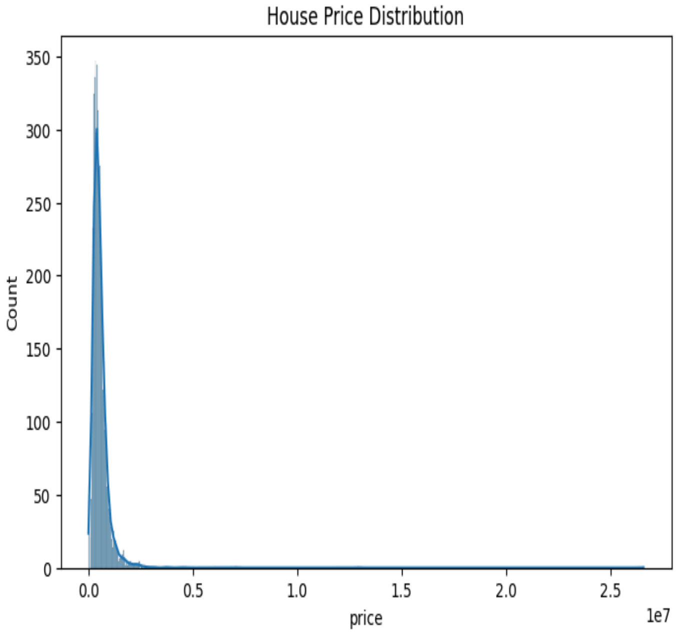
Target Variable:
- Price

Important Features:
- Bedrooms
- Bathrooms
- Square Foot Living Area
- Square Foot Lot
- Floors
- Waterfront
- View
- Condition
- Square Foot Above Ground
- Square Foot Basement
- Year Built
- Year Renovated
- Location Features

The dataset contains multiple numerical and categorical variables that influence house prices.

3. EXPLORATORY DATA ANALYSIS

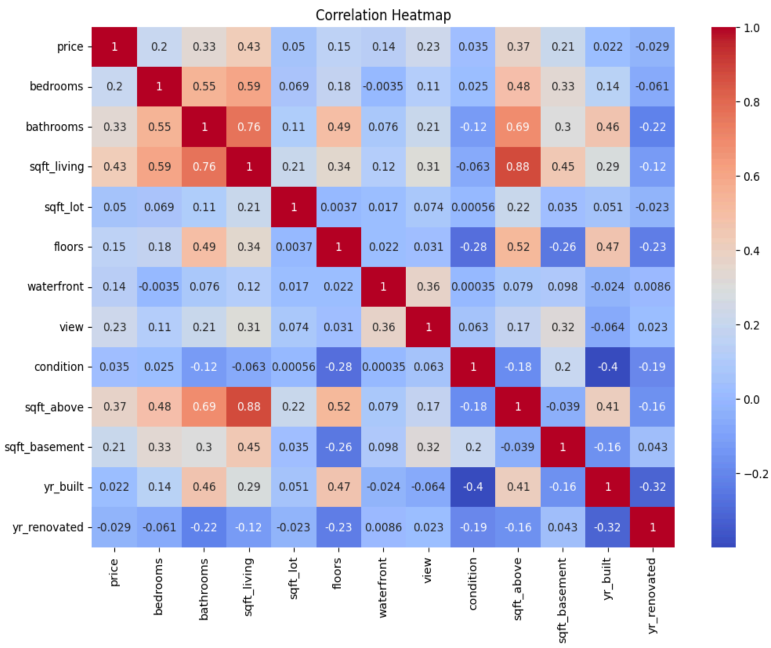
Figure 1: House Price Distribution



Observation:

The distribution of house prices is highly right-skewed. Most houses are concentrated within a lower price range, while a small number of properties have extremely high prices. This indicates the presence of outliers and justifies the use of log transformation during preprocessing.

Figure 2: Correlation Heatmap



Observation:

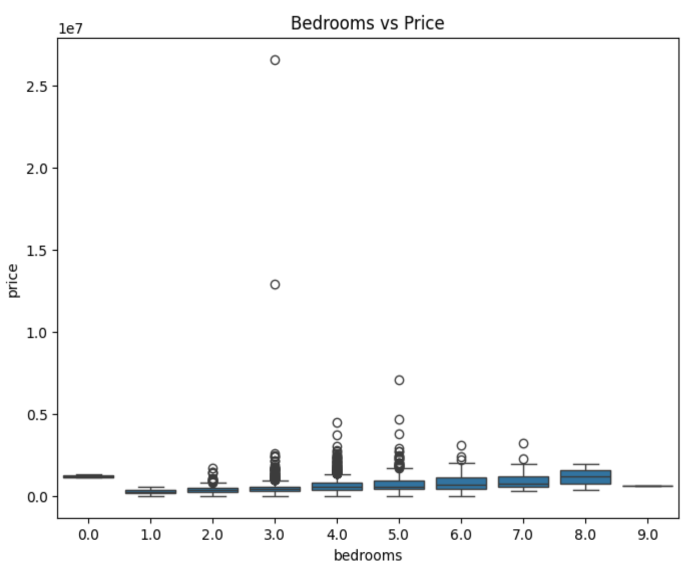
The heatmap shows relationships among the features and the target variable. Square footage related features such as sqft_living and sqft_above have the strongest positive correlation with house price. Bathrooms and bedrooms also show moderate positive correlations. Features like condition and year renovated have relatively weaker relationships with price.

Figure 3: Living Area vs Price

Observation:

The scatter plot reveals a positive relationship between living area and house price. As the living area increases, house prices generally increase as well. A few extreme outliers with exceptionally high prices can also be observed.

Figure 4: Bedrooms vs Price



Observation:

The boxplot indicates that houses with a greater number of bedrooms tend to have higher median prices. However, significant overlap exists among bedroom categories, suggesting that bedrooms alone are not sufficient to accurately predict house prices.

4. FEATURE ENGINEERING AND PREPROCESSING

The following preprocessing steps were performed:

• Date column converted into useful components such as sale year and sale month.
• Categorical variables encoded using Label Encoding.
• Log transformation applied to the target variable (price) to reduce skewness.
• Feature scaling performed using StandardScaler.
• Dataset divided into training and testing subsets using an 80:20 split.

5. MODEL TRAINING

Three regression algorithms were trained and compared:

1. Linear Regression
2. Random Forest Regressor
3. Gradient Boosting Regressor

These models were evaluated using regression metrics including Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2 Score.

6. MODEL EVALUATION

The trained models were compared based on prediction accuracy and error measurements.

Evaluation Metrics:

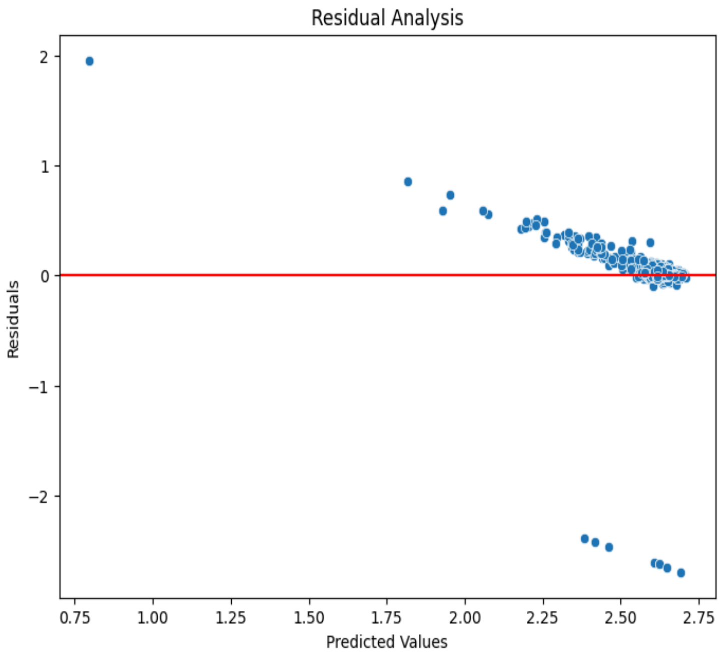
• Mean Absolute Error (MAE)
• Root Mean Squared Error (RMSE)
• R^2 Score

Observation:

The ensemble-based models (Random Forest and Gradient Boosting) are expected to perform better than Linear Regression because they can capture complex nonlinear relationships present in housing data.

7. RESIDUAL ANALYSIS

Figure 5: Residual Analysis



Observation:

The residual plot shows that most prediction errors are concentrated near the zero line, indicating that the model performs reasonably well on the majority of observations. A few large residuals are visible, which correspond to high-value outlier properties. Overall, the residual distribution suggests acceptable predictive performance.

8. MODEL SAVING

The best-performing model was saved using Joblib.

Saved Model File:

house_price_model.pkl

This model can be loaded later and used for predicting prices of unseen houses.

9. GOOGLE DRIVE MODEL LINK

Model File Link:

https://drive.google.com/drive/folders/1edjpzd_q4S87ZzZDjrAYF2g6vibpXrDt?usp=sharing

10. SUMMARY

This project focused on building a machine learning regression model to predict house prices using property-related features. The dataset contained information regarding house characteristics such as bedrooms, bathrooms, living area, lot size, floors, location attributes, and construction details. The primary objective was to understand the factors influencing house prices and develop an accurate predictive model.

Exploratory data analysis revealed that house prices were highly skewed, with a small number of luxury properties creating significant outliers. Correlation analysis indicated that living area, bathrooms, and above-ground square footage had the strongest relationships with house prices. Several visualizations were used to understand these relationships and identify important predictive features.

Data preprocessing included handling feature transformations, encoding categorical variables, applying logarithmic transformation to the target variable, and scaling numerical features. Three regression algorithms—Linear Regression, Random Forest Regressor, and Gradient Boosting Regressor—were trained and evaluated using MAE, RMSE, and R^2 Score.

Residual analysis demonstrated that the selected model produced reasonably accurate predictions for most properties, with only a few large errors associated with extreme-value houses. The best-performing model was saved as a deployment-ready artifact using Joblib.

This project successfully demonstrated the end-to-end machine learning workflow for regression problems, including data analysis, preprocessing, feature engineering, model development, evaluation, and model deployment preparation.